

CODE SPORT



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In this month's column, we celebrate the 10th anniversary of *LFY/OSFY* by looking back at how programming languages evolved over the last 10 years, and look ahead to what is in store for us over the next decade.

Welcome to a special edition of CodeSport. As you know, this month, we are celebrating the 10th anniversary of *LFY/OSFY*. This edition carries a number of articles featuring the 'Top 10 ...' in various domains, in celebration of our 10th anniversary. Celebrating 10 years is a grand milestone for *LFY/OSFY*. As we have journeyed through the last 10 years, the world of programming languages has witnessed many a change. In this month's column, we take a nostalgic look at how programming languages evolved over the past decade, and provide a peek into what the next 10 years may hold for us.

Programming languages over the past 10 years

Over the past 10 years, there have been many changes in programming languages, influenced by events in the software development ecosystem. We started 10 years back with the Dot Com boom and the bubble then burst; we went through the 'Web 2.0' revolution, and are currently being swept along by the 'Mobile Momentum' and 'Big Data' explosion.

Java ascended the throne of programming languages with the Dot Com boom and still continues to rule a significant part of the kingdom. The Web 2.0 revolution heralded the age of scripting languages. JavaScript became the de-facto client-side Web programming language. Python, Perl, PHP and Ruby came of age as mainstream development languages in the course of the Web 2.0 revolution. 'Mobile Momentum' gave us Objective-C and enshrined Java in Android. The Big Data explosion moved the spotlight from 'Code' to 'Data' and it became cool and nerdy for computer science graduates to re-invent

themselves as 'Data Scientists' instead of as 'geeky programmers'. Programmers learnt to develop skills in statistics, machine learning and data mining, as well as in traditional coding and testing techniques.

Given these developments in the software ecosystem, the popularity of different programming languages has fluctuated widely over the past 10 years. The well-known TIOBE programming community index (<http://www.tiobe.com>) measures the popularity of programming languages. It is enlightening to look at what languages were popular in 2003 and compare them with the TIOBE index in 2013. Look at Tables 1 and 2. Can you guess which table reflects the popular languages of 2012 and which shows 2003's Top 10?

It is pretty obvious that Table 2 deals with 2013; the dead giveaway is the presence of Objective-C as third most popular, propelled by the mobile app development focus on iOS. Table 1 gives the popular programming languages of 2003. It is interesting to compare the two tables that are 10 years apart. C, C++ and Java continue to rule the roost. Performance-intensive system software code still gets written in C or C++. The majority of application development is still in Java. Mobile application development on iOS and Android accounts for the popularity of Objective-C and Java. C# popularity can be attributed to Web/mobile application development on the Windows platform. Python's popularity and adoption has increased considerably over the last 10 years. If you are wondering where Ruby is in Table 2, Ruby was the 11th most popular programming language as of January 2013, and hence did not make it to Table 2. More detailed comparisons of